

CENTASTART-V

HIGHLY FLEXIBLE CLUTCHES & DYNAMIC COUPLING SYSTEMS

Product Overview: The CENTASTART-V series represents a premier selection of specialized flexible clutches and couplings engineered to optimize soft start-ups, minimize impact loads, and isolate heavy torsional resonances. Featuring integrated variable rubber qualities, it ensures secure switching speeds alongside excellent displacement absorption margins for heavy industrial drives and pump installations.

KEY PERFORMANCE ADVANTAGES

- **Controlled Switching Speed Profiles:** Engineered with explicit switching velocity limits (\$n_E\$) to ensure a smooth transition from idling phases to active torque transmission states.
- **Elite Shock & Vibration Dampening:** Features exceptional relative damping (\$\Psi\$) and dynamic torsional stiffness to counteract severe resonance accumulation cleanly.
- **Multidirectional Displacement Protection:** Isolates significant continuous axial, radial, and angular misalignments, minimizing fatigue across downstream shafts.

TECHNICAL SPECIFICATIONS & PERFORMANCE DATA (SIZES 80 - 6000)

Below is the calibrated engineering performance matrix for the **CENTASTART-V** series:

Size	Rubber Quality [Shore A]	Nominal Torque	Maximum Torque	Continuous Vibratory	Permissible Power Loss	Dynamic Torsional Stiffness	Relative Damping	Switching Speed	Max Speed	Permissible Displacement		
		T_{KN} [kNm]	T_{Kmax} [kNm]	T_{KV} [kNm]	P_{KV} [W]	C_{Tdyn} [kNm/rad]	Ψ	n_E [min ⁻¹]	n_{max} [min ⁻¹]	ΔK_a [mm]	ΔK_r [mm]	ΔK_v [°]
80	50	0,1	0,28	0,04	25	0,9	0,9	1100	5800	1	0,5	1
	60					1,5	1,5					
180	50	0,2	0,56	0,08	40	2	2	850 – 1100	5000	1	0,5	1
	60					3,4	3,4					
400	50	0,5	1,40	0,20	80	4,8	4,8	950	3800	1,5	0,5	1
	60					7,8	7,8	1000				
600	50	0,7	2,10	0,30	90	12	12	820	3800	1,5	0,5	1
	60					19	19	850				
900	50	1,1	3,15	0,45	120	10,5	10,5	870 – 1000	3000	1,5	0,5	1
	60					16	16	830 – 960				
1400	50	1,7	4,90	0,70	150	26,5	26,5	850 – 920	3000	1,5	1	1
	60					40	40	900 – 1000				
2500	50	3,0	8,75	1,25	200	43	43	750 – 850	2650	2	1	1
	60					77	77	800 – 850				
4000	50	5,0	12,50	2,00	240	75	75	720 – 900	2500	2	1	1
	60					120	120					

Size	Rubber Quality [Shore A]	Nominal Torque	Maximum Torque	Continuous Vibratory	Permissible Power Loss	Dynamic Torsional Stiffness	Relative Damping	Switching Speed	Max Speed	Permissible Displacement		
		T_{KN} [kNm]	T_{Kmax} [kNm]	T_{KW} [kNm]	P_{KV} [W]	C_{Tdyn} [kNm/rad]	ψ	n_E [min ⁻¹]	n_{max} [min ⁻¹]	ΔK_a [mm]	ΔK_y [mm]	ΔK_w [°]
6000	50	8,0	20,00	3,20	330	105	105	750 – 850	2300	2	1	1
	60					160	160					

* Values for idling speed and transmittable torque are available on request.
 ** Only valid for types VSS/VFS without bearing configuration.

PROCUREMENT, SIZING SELECTION & ENGINEERING SUPPORT (INDIA)

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